

1 Overview

Broadly speaking, the goal of this project is to develop interpretable machine learning methods by generating informative features from the data in the form of LLM queries and using the resulting answers as inputs for model training.

1.1 Binary Classification

As an initial problem we consider a binary classification task. In this case the idea is to prompt LLMs to identify queries that separates a small set of positive examples from a small set of negative examples. The resulting query is used as a structured feature and integrated as input into standard machine learning methods.

1.2 Interpretability Edge

By representing complex inputs through semantically meaningful features, the approach preserves the interpretability of models designed to be interpretable on structure data. In particular, the resulting query-based features can be used as inputs to models such as decision trees, naive Bayes classifiers, or Bayesian networks. This allows the final model to remain transparent and interpretable while still leveraging information contained in unstructured data sources.

Tree Example: We consider two possible constructions of the feature space. One approach is to generate a global feature space consisting of a large shared set of queries that are available to all splits in the tree. Alternatively, one can construct a local feature space, where the queries considered at each node are generated using a subset of positive and negative samples that reach that node. In this case, the candidate queries are tailored to the local classification task defined by the data partition induced by the tree.

1.3 Decorrelation Edge

The intrinsic randomness in the query generation process can help decorrelate models in ensemble methods. Because queries are generated rather than selected from a fixed set of variables, the space of potential features is substantially larger. For example, in random forests the queries used across different trees can be made highly diverse, leading to weaker correlations among trees and potentially improved ensemble performance.

1.4 Contrastive Learning Edge

The framework enables a more contrastive form of learning by allowing the subset of data points used to generate queries to be selected with complete flexibility. For example, in boosting, queries can be generated explicitly to distinguish examples that previous trees classified incorrectly.

1.5 Types of Queries

An important question is whether the LLM should generate only presence queries, such as: “Does the patient have a diagnosis of diabetes in their record?” or whether it should also generate inference queries, such as: “Is this patient likely to be diabetic?” If inference queries are allowed, additional challenges arise, including more uncertain or less consistent answers and the potential creation of proxies for the outcome variable. One possible approach is to use the proposed framework recursively to analyze why the LLM made a given inference.

1.6 Variable Importance

For the case of global feature space (a shared large set of queries across all trees) we can apply standard variable importance methods. For the case of local query features, existing approaches cannot be immediately applied. Some possibilities for adapting variable importance measures in this case are:

Gini importance: cluster splits and collect importance for equivalent splits.

Permutation importance: ask an LLM to de-identify the data with respect to a given feature.

1.7 Extensions

Possible extensions beyond the initial setting include:

- Multi-class classification
- Regression
- Clustering
- Bayesian Network Structure Learning
- Anomaly Detection
- Policy Learning (e.g. Prescriptive Trees)
- Causal Inference

1.8 Revisiting Machine Learning Challenges

This framework raises several classical machine learning challenges that must be addressed:

- Overfitting
- Instability
- Creation of variables that act as proxies for the outcome
- Uncertainty and consistency in the answers generated by the LLM
- Propagation of biases